Decoding the IPL

Can Data Predict the Next Champion?

Analytics Induction Programme · IPL Dataset 2008–2022

1. Introduction

This report presents the complete data analytics pipeline applied to the IPL Complete Dataset (2008–2022). The project addresses the question: given the venue, competing teams, and toss decision at the start of a match, can a logistic regression classifier predict whether the team batting first will win?

The dataset has two files — a matches file (Dataset1.csv: 950 rows, 20 columns) and a ball-by-ball file (225,954 rows, 17 columns). After removing 4 no-result matches, 946 valid matches were used across all analyses.

2. Data Loading and Cleaning

Both datasets were loaded using pd.read_csv() and inspected using head(), info(), and describe(). The matches dataset has 950 rows. Key observations: the City column has 51 missing values, Margin has 18 missing values, and the method column is mostly empty (only 19 non-null). These columns are not used in the model so no imputation was needed.

Cleaning steps performed: (1) Removed 4 matches where WonBy = "NoResults". (2) Normalised season labels — "2007/08" → "2008", "2009/10" → "2010", "2020/21" → "2020". (3) Standardised venue names to remove duplicates caused by different naming conventions (e.g. "Eden Gardens, Kolkata" → "Eden Gardens"). After cleaning, 946 valid matches remained with no null values in the key columns (Team1, Team2, Venue, TossWinner, TossDecision, WinningTeam).

The target variable **bat_first_won** was engineered: if TossDecision = "bat", then bat_first_won = 1 if TossWinner also won the match, else 0. If TossDecision = "field", then bat_first_won = 1 if TossWinner lost the match (since the other team batted first). The bat-first win rate across all 946 matches is **44.926%**, meaning the chasing team wins slightly more often.

Innings totals were computed from the ball-by-ball data by summing total_run per match-innings pair. After filtering to innings 1 and 2 only (removing super overs), 1,898 innings records remained. Score statistics: mean = 155.85, std = 31.21, min = 2, max = 263.

3. Statistical Visualisation

Eight plots were created covering the required chart types. All plots were saved as PNG files at dpi=300.

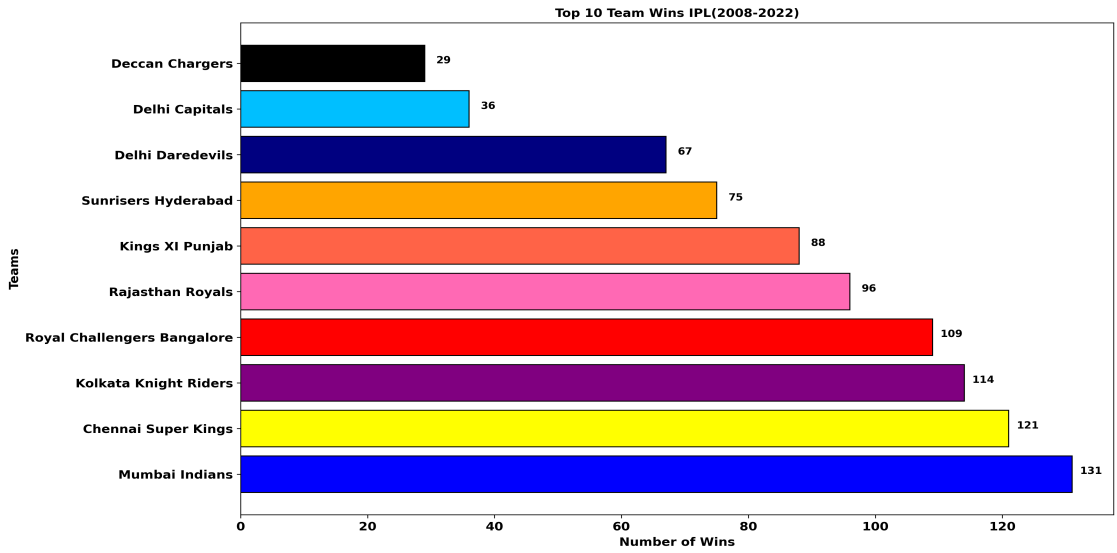

Figure 1: Top 10 teams by total wins in IPL 2008–2022 (bar chart).

Mumbai Indians lead with the most wins, followed by Chennai Super Kings and Kolkata Knight Riders. This reflects their consistent performance over 15 seasons.

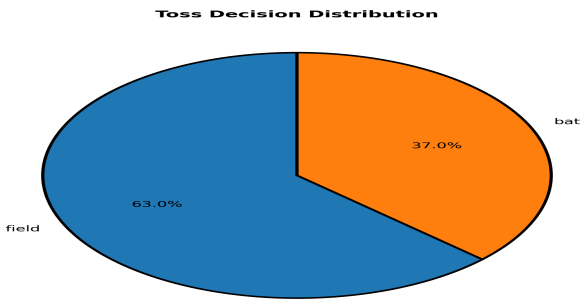

Figure 2: Toss decision distribution — field vs bat (pie chart).

About 63.2% of toss winners chose to field first, suggesting a general preference for chasing in T20 cricket, likely due to dew factor in evening matches.

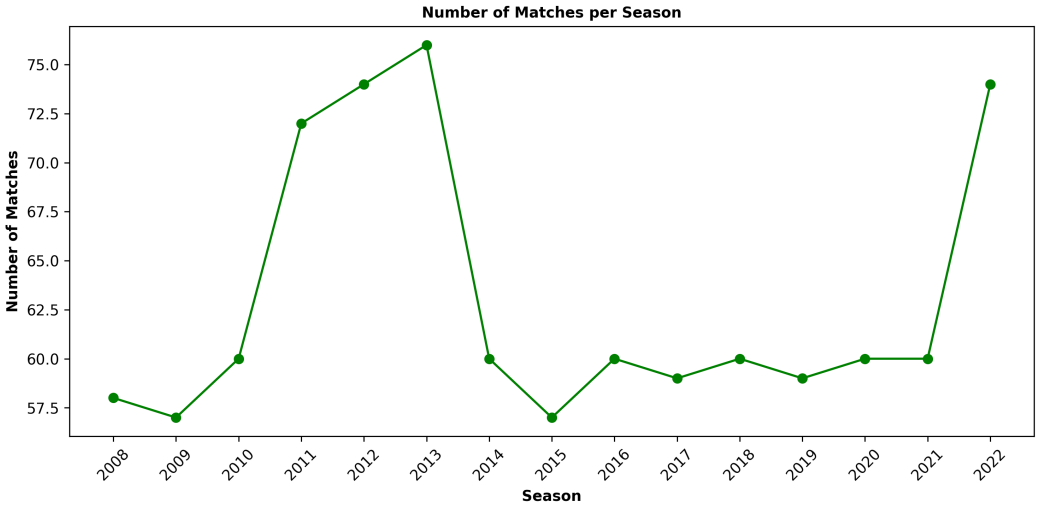

Figure 3: Number of matches played per IPL season (line chart).

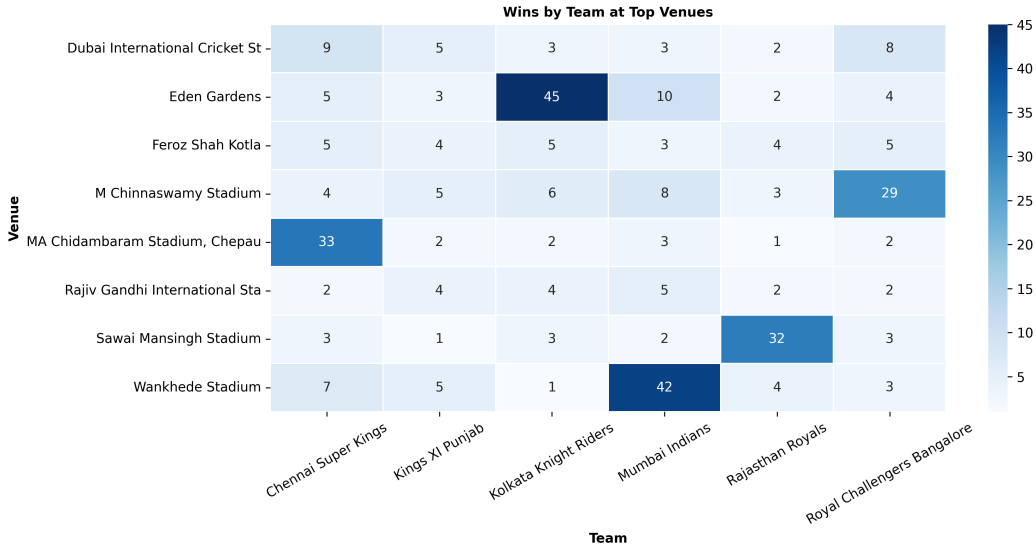

Figure 4: Wins by team at the top 8 venues — top 6 teams (heatmap).

The heatmap shows clear home-ground advantages. Mumbai Indians dominate at Wankhede Stadium and Chennai Super Kings at MA Chidambaram Stadium, Chepauk.

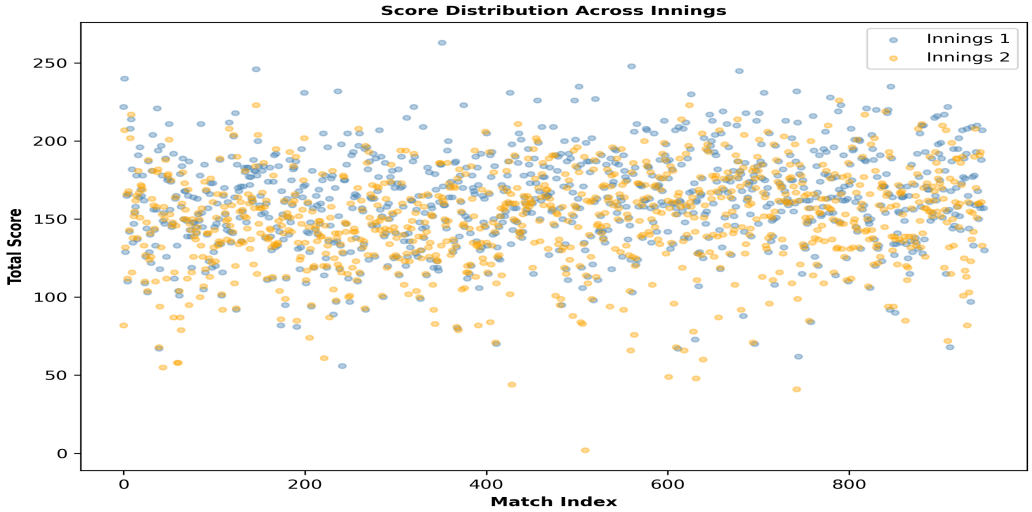

Figure 5: Scatter plot of innings scores — innings 1 vs innings 2.

Both innings follow a similar distribution centred around 155 runs. There is no visible systematic difference between first and second innings scores.

4. Outlier Detection

Two methods were used to identify statistically extreme innings scores from the 1,898 innings records.

IQR Method

Q1 = 137.0, Q3 = 176.75, IQR = 39.75. The Tukey fences are: Lower fence = 77.4, Upper fence = 236.4. Any innings score below 77.4 or above 236.4 is flagged. This method detected **32 outlier innings**.

Z-Score Method

For each innings score, a Z-score was computed as (score – mean) / std. Innings with |z| > 3 were flagged. This method detected **13 outlier innings**. All 13 were already caught by the IQR method.

Comparison

The IQR method is more sensitive — it catches 32 outliers including moderate extremes in the 60–76 run range. The Z-score method is stricter and only flags the most extreme values (scores of 2, 41, 44, 48, 49, 55, 56, 58, 60, 61, 62, and 263). The Z-score outliers are a subset of the IQR outliers.

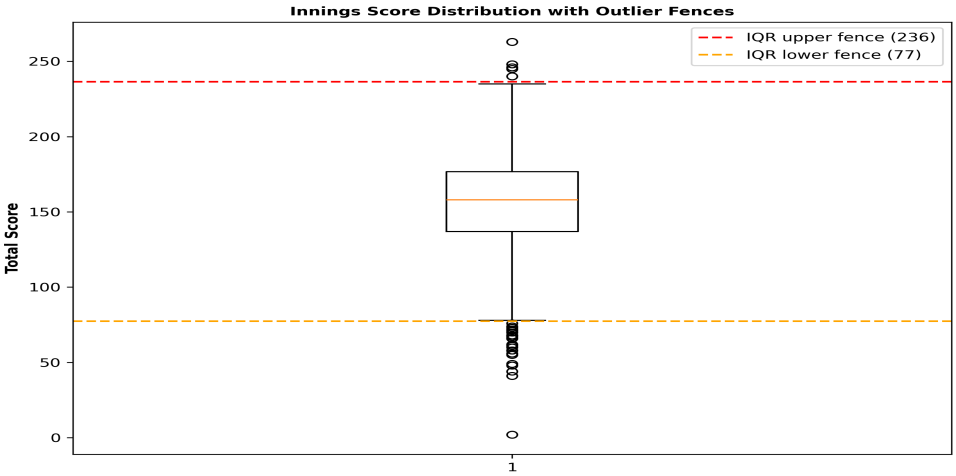

Figure 6: Box plot showing innings score distribution with IQR fences.

5. Feature Engineering and Model Construction

The following features were prepared from the matches dataset:

Feature	Type	How it was created
team1_won_toss	Binary (0/1)	1 if TossWinner == Team1, else 0
TossDecision	Encoded (0/1)	LabelEncoder applied to "bat"/"field"
Venue (one-hot)	Binary columns	pd.get_dummies on Venue (37 columns)
Team1 (one-hot)	Binary columns	pd.get_dummies on Team1 (17 columns)
Team2 (one-hot)	Binary columns	pd.get_dummies on Team2 (17 columns)

The final feature matrix had 946 rows and 73 columns. The target variable was bat_first_won (0 or 1). A stratified train-test split was applied with test_size=0.2 and random_state=42, giving 756 training samples and 190 test samples. A Logistic Regression model was trained with max_iter=1000, random_state=42.

6. Model Evaluation

D3 — Metric Summary Table

Metric	Training Set	Test Set
Accuracy	0.6376	0.5368
Precision	0.6387	0.4789
Recall	0.4471	0.4
F1-Score	0.526	0.4359
ROC-AUC	0.6811	0.5393

The model achieved a test accuracy of 53.68% and a test AUC of 0.539, only marginally above random guessing (50% / AUC=0.5). The training accuracy of 63.76% is noticeably higher than test accuracy, suggesting some overfitting — the model memorises training patterns that do not fully generalise to unseen matches. This is expected given that the feature set (team names, venue, toss) has limited true predictive power.

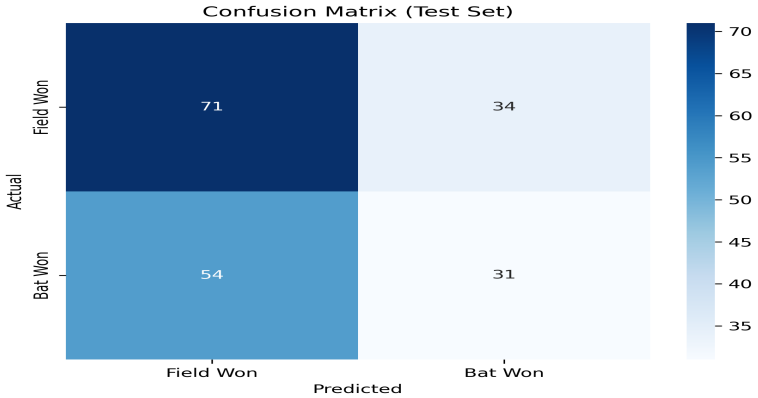

Figure 7: Confusion matrix on the test set (190 matches).

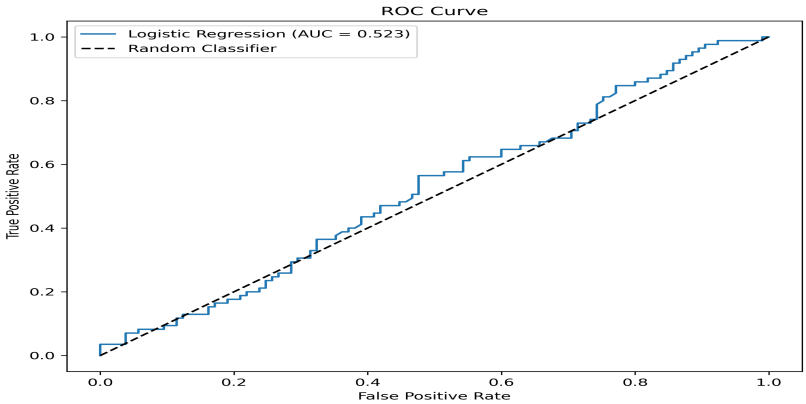

Figure 8: ROC curve — test set AUC = 0.539.

7. Chi-Square Hypothesis Test

A Chi-Square test of independence was conducted to determine whether the toss decision (bat vs field) is statistically associated with the match outcome (bat_first_won), at significance level $\alpha = 0.05$.

Hypotheses:

- H0: Toss decision and bat_first_won are independent.
- H1: Toss decision and bat_first_won are associated.

Contingency Table:

TossDecision	bat_first_won = 0	bat_first_won = 1
bat	191	159
field	330	266

D4 — Statistical Test Result:

Chi-Square Statistic	Degrees of Freedom	p-value	Decision ($\alpha = 0.05$)
0.0291	1	0.8647	Fail to Reject H0

Since $p = 0.8647 >> 0.05$, we fail to reject the null hypothesis. Toss decision is NOT significantly associated with match outcome. Choosing to bat or field after winning the toss does not give a team a statistically measurable advantage based on 15 seasons of IPL data.

8. Conclusions

Key findings from this analysis:

- Batting-first teams win only 44.9% of matches — the chasing team has a slight advantage.
- Mumbai Indians lead all-time wins. About 63.2% of captains choose to field after winning the toss.
- IQR detected 32 outlier innings; Z-score detected 13. IQR is more sensitive, Z-score is stricter.
- Logistic Regression: test accuracy = 53.68%, test AUC = 0.539 — limited predictive power from pre-match features.
- Chi-square test ($\chi^2 = 0.0291$, $p = 0.8647$): no significant association between toss decision and outcome.
- Future improvement: add rolling form features, head-to-head records, and try Random Forest.